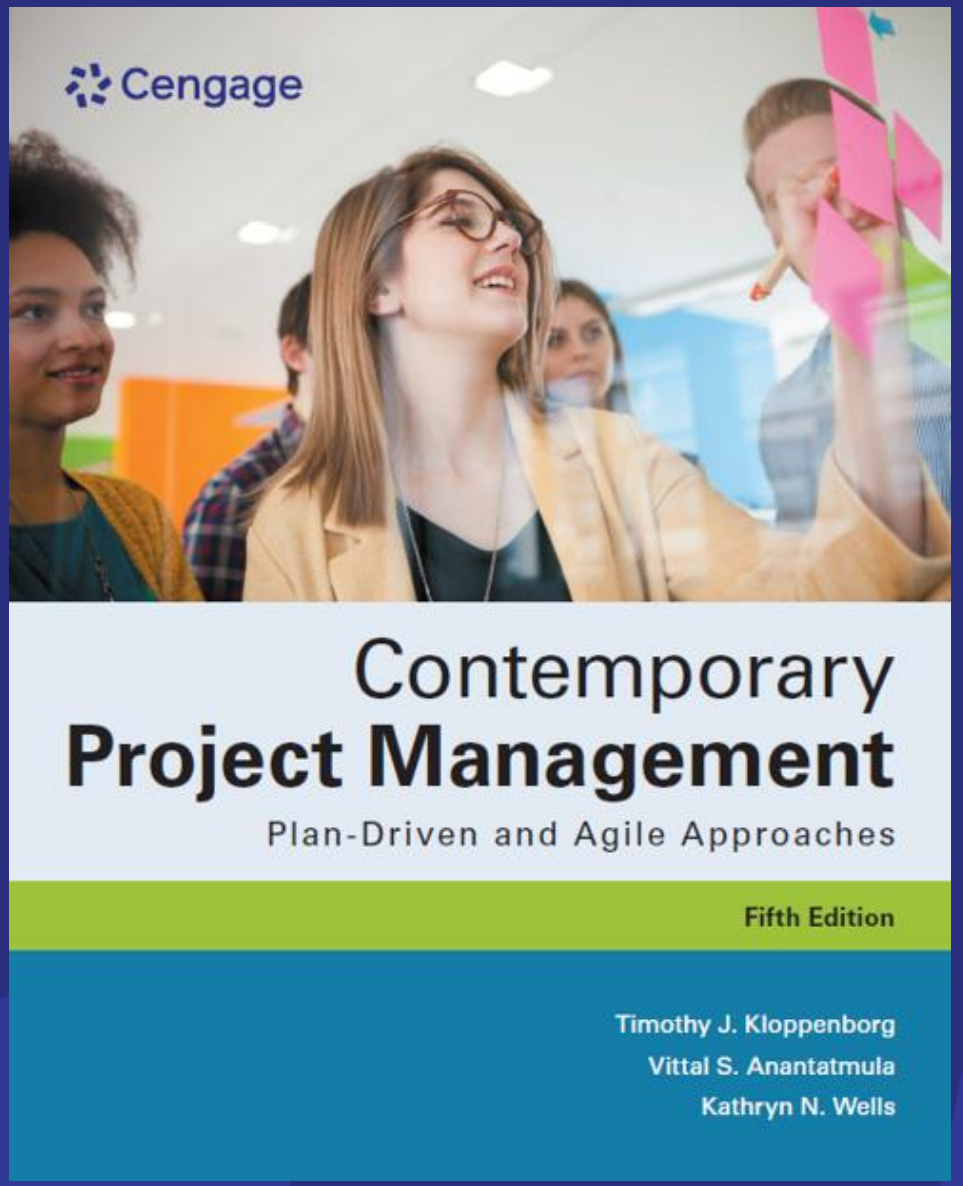


Chapter 11

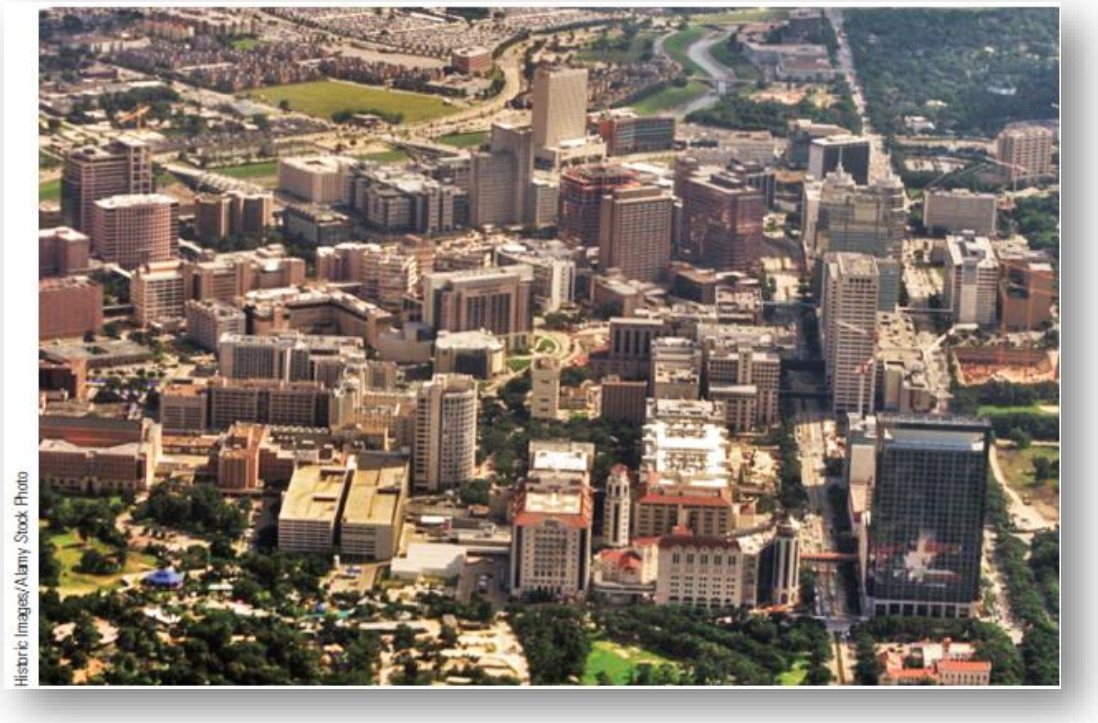
Project Uncertainty Planning

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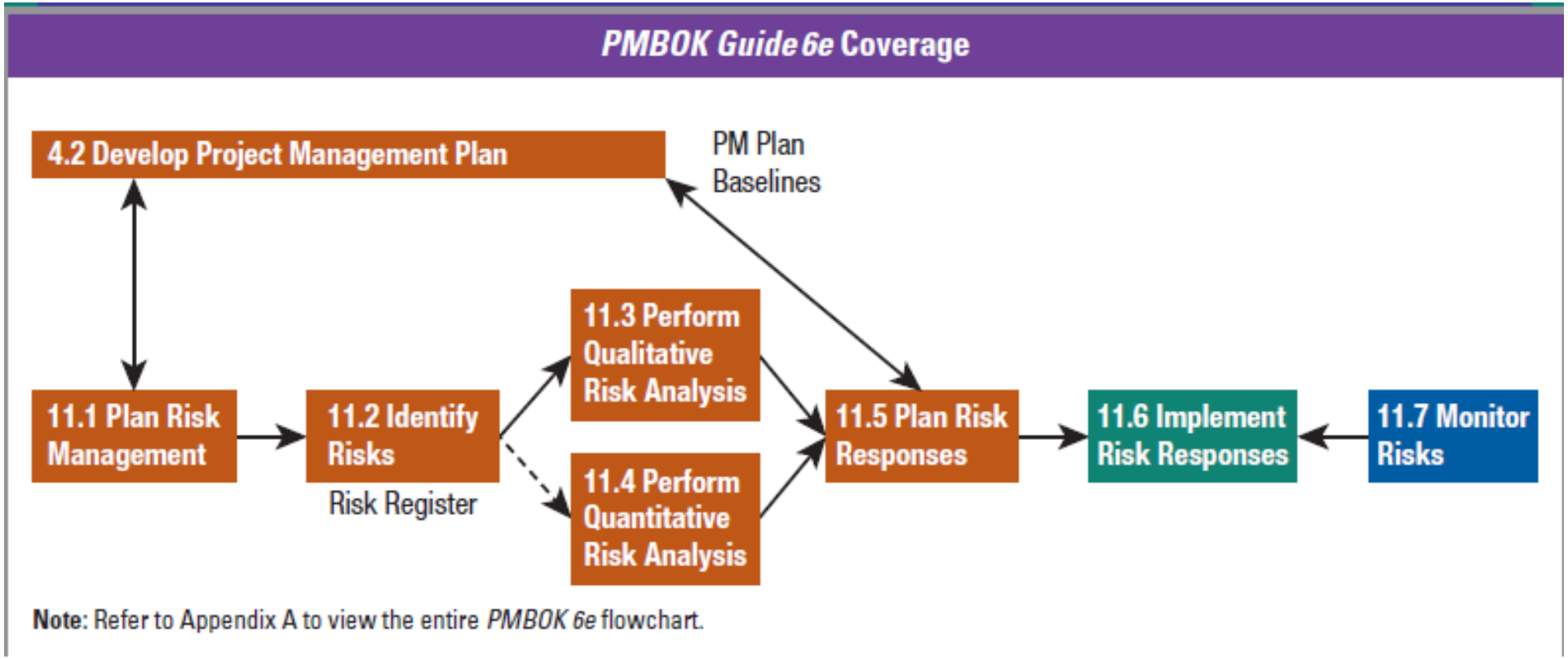
Icebreaker

What are two things you learned from Budgeting Projects?



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Project Uncertainty Planning



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Project Risk Planning

Overview of the Book
11.1 Plan Risk Management
11.2 Identify Risks
11.3 Risk Analysis
11.4 Plan Risk Responses
11.5 *PMBOK Guide 7e*
11.6 Agile Projects
Chapter Summary

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Core Objectives

- **11-1** Describe the processes of identifying risks, analyzing risks, and developing response plans for identified risks.
- **11-2** Identify and classify risks for a project and populate a risk register.
- **11-3** Describe various risk assessment techniques and state when each technique is appropriate to use.
- **11-4** Prioritize each risk on a project using an appropriate assessment technique and develop and defend at least one strategy for each of the high-priority risks.
- **11-5** Compare and contrast the various strategies for dealing with risks.
- **11-6** Select and utilize an appropriate quantitative risk analysis tool if qualitative risk analysis is not sufficient.
- **11-7** Determine the propensity of a key decision-maker to accept risk and use that knowledge to strategize about.

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Agile Objectives

- **11-8** Explain how risk is generally reduced or mitigated using Agile methods.
- **11-9** Plan and conduct a risk-based spike to prove your approach to your product owner.

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11.1 Plan Risk Management (1 of 3)

11.1a Roles and Responsibilities
11.1b Categories and Definitions

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11.1 Plan Risk Management (2 of 3)

Plan Risk Management is “the process of defining how to conduct risk management activities for a project.”

- Practice Standard for Project Risk Management (PMI)

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11.1 Plan Risk Management (3 of 3)

What constitutes project success?

- Meeting various agreements *(i.e., contracts)*
- Did project meet customers' expectations?
- Did project help the performing organization?
  - New technologies
  - Employee growth

Meeting Agreements

Cost, schedule, and specifications met

Customer's Success

Needs met, deliverables used, customer satisfied

Performing Organization's Success

Market share, new products, new technology

Project Team's Success

Loyalty, development, satisfaction

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Exhibit 11.2

Specific Project Stakeholder Priorities

	Improve	Keep	
Scope		X	
Quality		X	
Time			≤ 1 month to save $5,000
Cost	Want to save		
Contribution to Organization		X	
Contribution to Society		X	

Source: Adapted from Kloppenborg, Timothy J. and Joseph A. Petrick, *Managing Project Quality* (Vienna, VA: Management Concepts, Inc., 2002): 46.

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Risk Management Planning and Stakeholder Priorities

- Understand what the project plan calls for.
- Understand area the most important stakeholders like to improve.
- Understand where stakeholders are willing to sacrifice to enable improvements.

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Understanding the Project Risk (1 of 2)

Risk management includes guidance on how to perform three risk management activities:

1. Decide what level of risk premium to charge for the project. The team must rate factors such as project size, complexity, technology, and type. The combined ratings dictate that a risk premium of zero, ten, or twenty percent be added to the estimated project cost or, for very risky projects, that executive approval is mandated.
2. Mitigate risk external to the firm through contract clauses and risk internal to the firm through agreements.
3. Manage the risk very carefully through specifically designed weekly conference call meetings and reports.

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Understanding the Project Risk (2 of 2)

Project Risk is anything that may impact the project team's ability to achieve the general project success measures and the specific project stakeholders' priorities.

Threat is "a risk that will have a negative impact on the project objective if it occurs."

- Practice Standard for Project Risk Management (PMI)

Opportunity is "a risk that would have a positive effect on one or more project objectives."

- Practice Standard for Project Risk Management (PMI)

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Plan Risk Management

Risk Management Plan is part of the Project Management Plan that describes how risk management activities will be prioritized, monitored, planned and performed; also used for communicating with project stakeholders and for follow-up analysis.

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Exhibit 11.3

Risk Management Plan Guidance for an IT Consulting Company

Risk management includes guidance on how to perform three risk management activities:

1. Decide what level of risk premium to charge for the project. The team must rate factors such as project size, complexity, technology, and type. The combined ratings dictate that a risk premium of zero, ten, or twenty percent be added to the estimated project cost or, for very risky projects, that executive approval is mandated.
2. Mitigate risk external to the firm through contract clauses and risk internal to the firm through agreements.
3. Manage the risk very carefully through specifically designed weekly conference call meetings and reports.

Source: Rachana Thariani, PMP

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11.1a Roles and Responsibilities

- Encourage wide participation in risk management activities.
- More perspectives considered leads to more risks uncovered.
- Participation encourages buy-in to a risk management approach.
- Identify who is responsible for each risk.

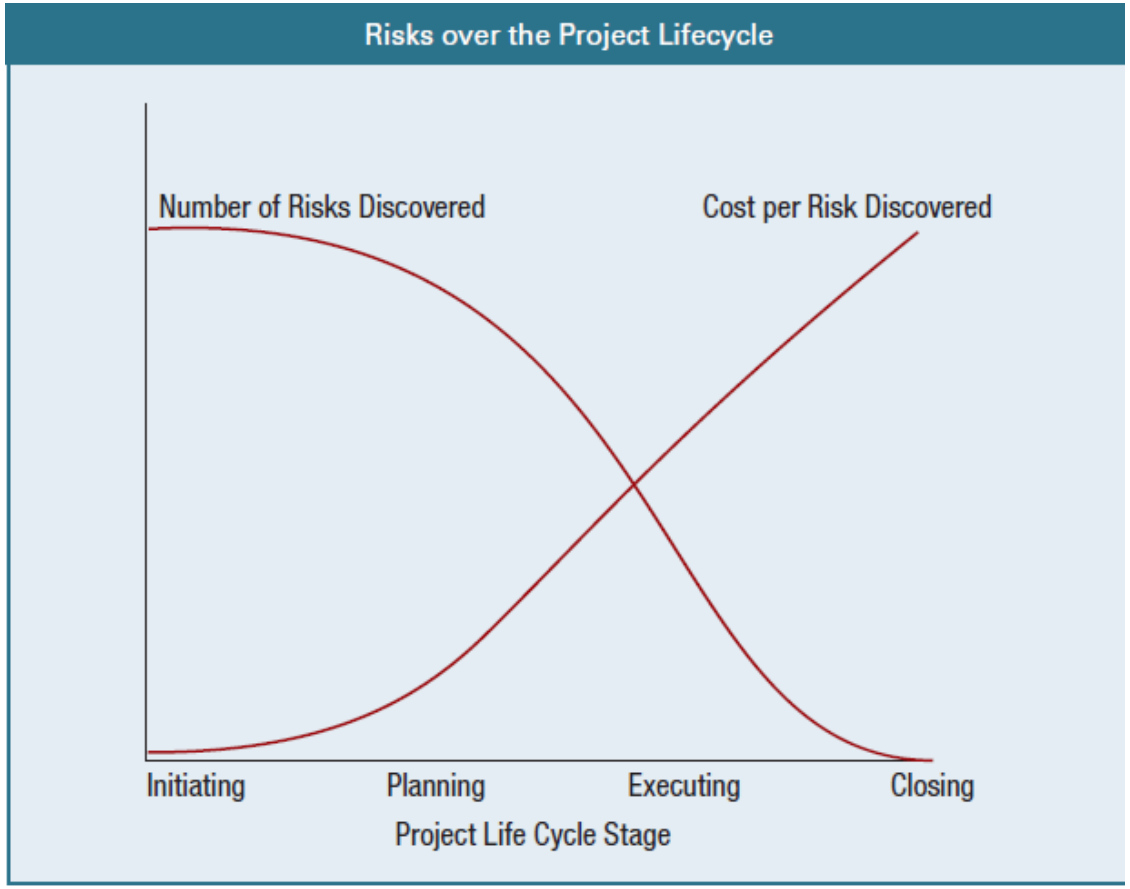
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11.1b Categories and Definitions (1 of 2)

- More project risks generally uncovered early in the life of a project.
- The cost **per risk** discovered early is less.
- Risks discovered late in a project can be expensive.
- Categorizing risks often leads to more thorough risk identification.
- Classify by:
  - Stage of project lifecycle.
  - Project objective that would be impacted.
  - External versus internal to project.
  - What is known versus unknown *(uncertainty)*.

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Exhibit 11.4



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11.1b Categories and Definitions (2 of 2)

- A “known-known” can be planned and managed with certainty.
- “Known-unknowns” can be identified and may or may not happen.
- “Unknown-unknowns” are true uncertainties.

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Exhibit 11.5

The Fourteen Most Important Risks in Panama Canal Expansion

Changes in design and quantities	Extreme bad weather
General inflation	Inadequate claims administration
Ineffective contracting process	Inefficient planning
Insufficient revenues	Lack of controls
Lack of skilled and local labor	Local labor strikes
Material, equipment, and labor cost	Organizational risks
Owner-driven changes	Referendum delays

Source: Alarcon, Luis F., et al., "Risk Planning and Management for the Panama Canal Expansion Program," *Journal of Construction Engineering and Management* (October 2011): 762–770.

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Exhibit 11.6

Top Risks in Each Factor for International Projects

Resources	Regulations	Crisis	Insurance	Science/Health	Digital
• Inflation/price Instability • Commodity price shocks • Supply chain disruptions • Natural resource shortages • Environmental changes • Talent/skill shortage	• Different and changing regulations • Interstate relations failure • Collapse of multilateral institutions • Erosion of social cohesion • Political power transitions	• Economic crisis • Debt crisis • Asset bubble burst • Collapse or lack of social security Systems • Terrorist strikes • State collapse • Youth disillusionment	• Differences in risk appetite or aversion • Insurance market challenges • Collapse of an important industry • Safety issues	• Backlash against science • Infectious disease • Vaccine challenges	• Failure of technology governance • IT infrastructure breakdown • Increased cyberthreats • Digital power concentration • Digital inequality • Private information compromised

Source: Adapted from Four Globalization Risks Impacting International Projects, Visualized: A Global Risk Assessment of 2021 and Beyond, Global Risks Perception Initiative—2021 Report, Five Top 2021 Trends Posing Risks to Construction Industry and Steffey, Robert W., and Vittal S. Anantatmula, "International Projects Proposal Analysis: Risk Assessment Using Radial Maps," *Project Management Journal* (April 2011): 62–70.

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Exhibit 11.7

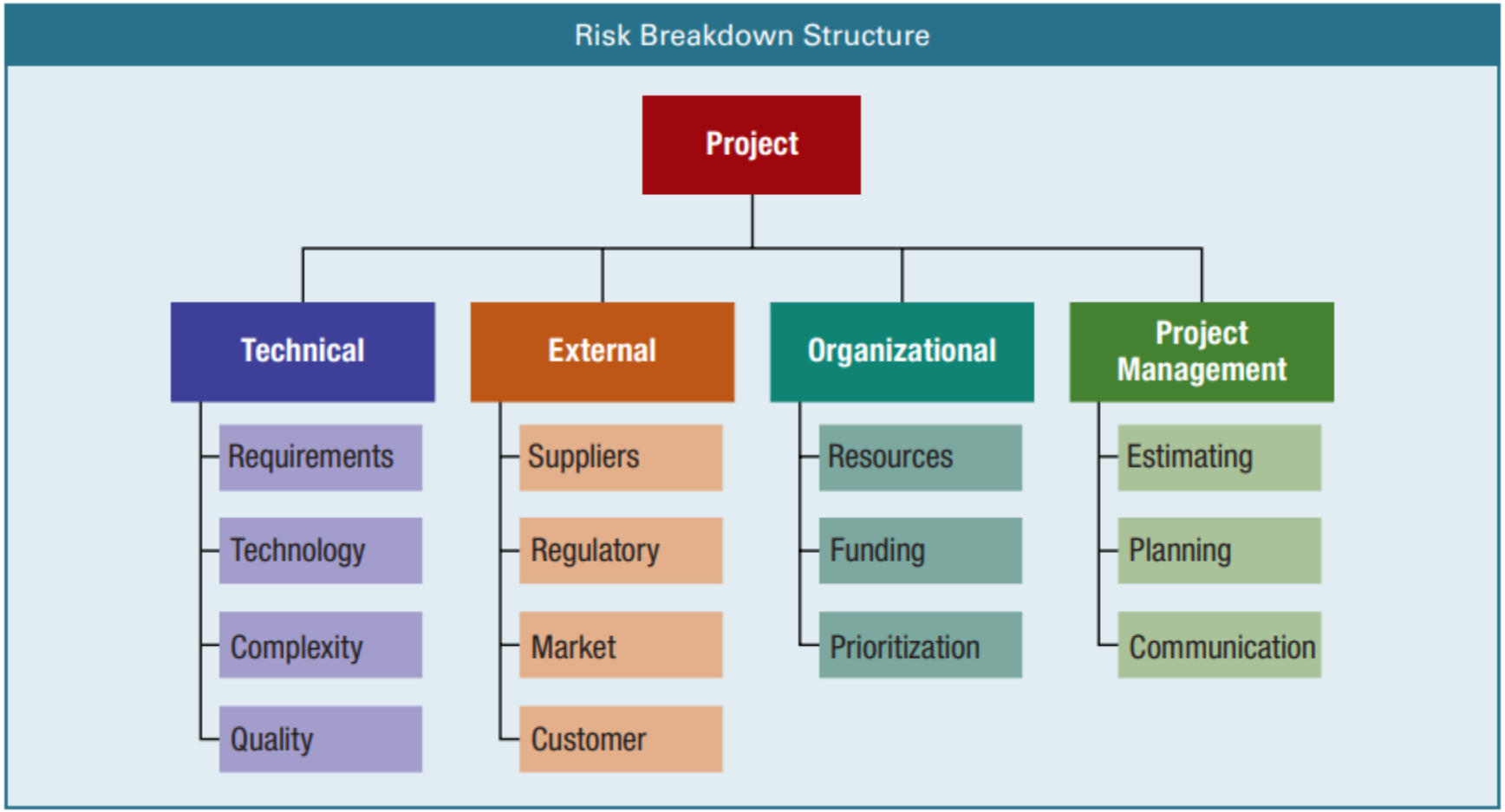
Top Risks in Each Factor for Software Projects

Execution Management	**User Coordination**
Configuration management system	User evaluation of progress
Formality of status reports	User understanding of complexity
Specification approval process	Care in user manual preparation
Post-project audits	Coordination with user
Regularity of technical reviews	Informal communication channels
Human Resource Management	**Project Planning**
Flexibility of working hours	Frequency of software reuse
Individual performance incentives	Planning tools used
Technical assistance availability	Minimum cost software design
Recognition for extra work	Removal of unnecessary requirements
Enforced attendance system	Individual accountability

Source: Thomas, Sam, and M. Bhasi, "A Structural Model for Software Project Risk Management," Journal of Management (March 2011): 71–84.

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Exhibit 11.8



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11.2 Identify Risks (1 of 2)

11.2a Information Gathering
11.2b Reviews
11.2c Understanding Relationships
11.2d Risk Register

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11.2 Identify Risks (2 of 2)

Identify Risks is “the process of determining which risks might affect the project and documenting their characteristics.”

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11.2a Information Gathering (1 of 2)

- Brainstorming activity considering “what could go wrong” can be used.
- Variations and extensions of possible risks can help identify additional risks.
- Project team members may interview stakeholders.
- **SWOT** analysis, Delphi technique can also be used.

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11.2a Information Gathering (2 of 2)

SWOT Analysis is a detailed analysis of the project's strengths, weaknesses, opportunities, and threats.

Delphi Technique is an information-gathering technique whose goal is to reach consensus among a group of experts; involves soliciting anonymous expert opinions, summarizing them, and sending them back to the experts for further comments and improvements.

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Exhibit 11.9

Project Risk Reviews

Type of Review	Question
Charter	Is there clarity and common understanding in each section? Are the sections consistent with each other?
Stakeholder register	What could upset any of them?
Communication plan	Where could poor communications cause trouble (or threaten project goals)?
Assumptions	Can you verify that each assumption is correct?
Constraints	How does each constraint make the project more difficult?
WBS	What risks can you find for each WBS item? What can go wrong with each WBS work element?
Schedule	What milestones and other handoff points might be troublesome?
Resource demands	At what points are certain people overloaded?
Touchpoints	What difficulties may arise when some project work is handed off from one person to another?
Literature	What problems and opportunities have been published concerning similar projects?
Previous projects	What projects and opportunities have similar projects in your own organization experienced? Can we gain some insights from the lessons learned database?
Peers	Can your peers identify any additional risks?
Senior management	Can senior management identify any additional risks?

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11.2c Understanding Relationships (1 of 2)

Root Cause Analysis is “an analytical technique to ascertain the fundamental or causal reason(s) that affect one or more variances, defects, or risks.”

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11.2c Understanding Relationships (2 of 2)

Trigger Condition is “a circumstance under which a risk strategy or risk action will be invoked.”

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11.2d Risk Register

Risk Register is “the document containing the results of the qualitative risk analysis, quantitative risk analysis, and risk response planning. It identifies all identified risks, including description, category, cause, probability of occurring, impact(s) on objectives, proposed responses, owners, and current status.”

- Practice Standard for Project Risk Management (PMI)

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Exhibit 11.10

Partial Risk Register

Risk Description (Event)	Impact	Category	Probability	Impact	Score	Mitigation Strategy-Resolution
Incomplete requirements were identified in the RFP and Exhibits (see Risk 7).	Greater possibility of gaps in functionality. Greater possibility of missing State-specific functionality. Greater possibility of "Scope Creep." Greater possibility of delay in finalizing requirements. Greater possibility of rework in subsequent phases.	Business Requirements	5	4s	20	MAXIMUS will begin conducting the detailed BA sessions 09/20/2012. Additional requirements will be gathered in those sessions and documented in subsequent versions of the Requirements Validation Documentation. A schedule of future Business Architecture and Technical sessions is being developed. State will provide closure and decisions regarding requirements and system scope.
Since there are various vendor products (IBM/Curam, Connecture), each with its own rules engines, it is not clear which rules engine takes precedence.	Potential duplication of rules or conflicting rules that lead to different outcomes.	Technology	3	4	12	Engage point will explain how to mitigate this risk. (See Risk Response Plan for resolution.)
Difficulty integrating to state's end-to-end infrastructure.	Potential difficulty integrating new technology into existing infrastructure.	Technology	4	4	16	Work with the state to define infrastructure requirements and ensure that the necessary information is being provided to the MN-IT staff.
Going through a hierarchical reporting structure will impact real-time decision making.	Potential bottlenecks in document reviews and decision making may affect task completion according to the project schedule.	Communications	4	4	16	Identifying a point-of-contact for each functional area from vendor and state to eliminate bottlenecks.
State functional POCs may have competing priorities that will hinder their ability to respond promptly.	Secondary risk—related to Risk 6.	Communications	4	4	16	Identify multiple points-of-contact for each functional area from vendor and state to eliminate bottlenecks.
Delays in procurement process may negatively impact the project schedule.	Inability to acquire resources promptly may negatively impact-related activities in the project schedule.	Procurement	2	5	10	Add lead time as early as possible. Evaluate procurement requirements during the change order process. Make sure commerce procurement staff are engaged in the PO development process.

Source: http://mn.gov/hix/images/BC9-l-ITAttachmentN.pdf, accessed April 26, 2013.

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11.3 Risk Analysis

11.3a Perform Qualitative Risk Analysis
11.3b Perform Quantitative Risk Analysis
11.3c Risk Register Updates

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11.3a Perform Qualitative Risk Analysis

Perform Qualitative Risk Analysis is “the process of prioritizing risks for further analysis or action by assessing and combining their probability of occurrence and impact.”

- Practice Standard for Project Risk Management (PMI)

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Exhibit 11.11

Qualitative Risk Assessment

Probability		**Impact**			
	Insignificant (1)	**Minor (2)**	**Moderate (3)**	**Major (4)**	**Catastrophic (5)**
Almost certain (>90% chance)	High	High	Extreme	Extreme	Extreme
Likely (50–90%)	Moderate	High	High	Extreme	Extreme
Moderate (10–50%)	Low	Moderate	High	Extreme	Extreme
Unlikely (3–10%)	Low	Low	Moderate	High	Extreme
Rare (<3%)	Low	Low	Moderate	High	High

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11.3a Qualitative Risk Analysis

Cause-and-Effect Relationships

- One way to assess magnitude of effect is to understand cause.
- Part of root cause analysis.
- Change the effect by changing the underlying cause.
- Use a cause-and-effect diagram.

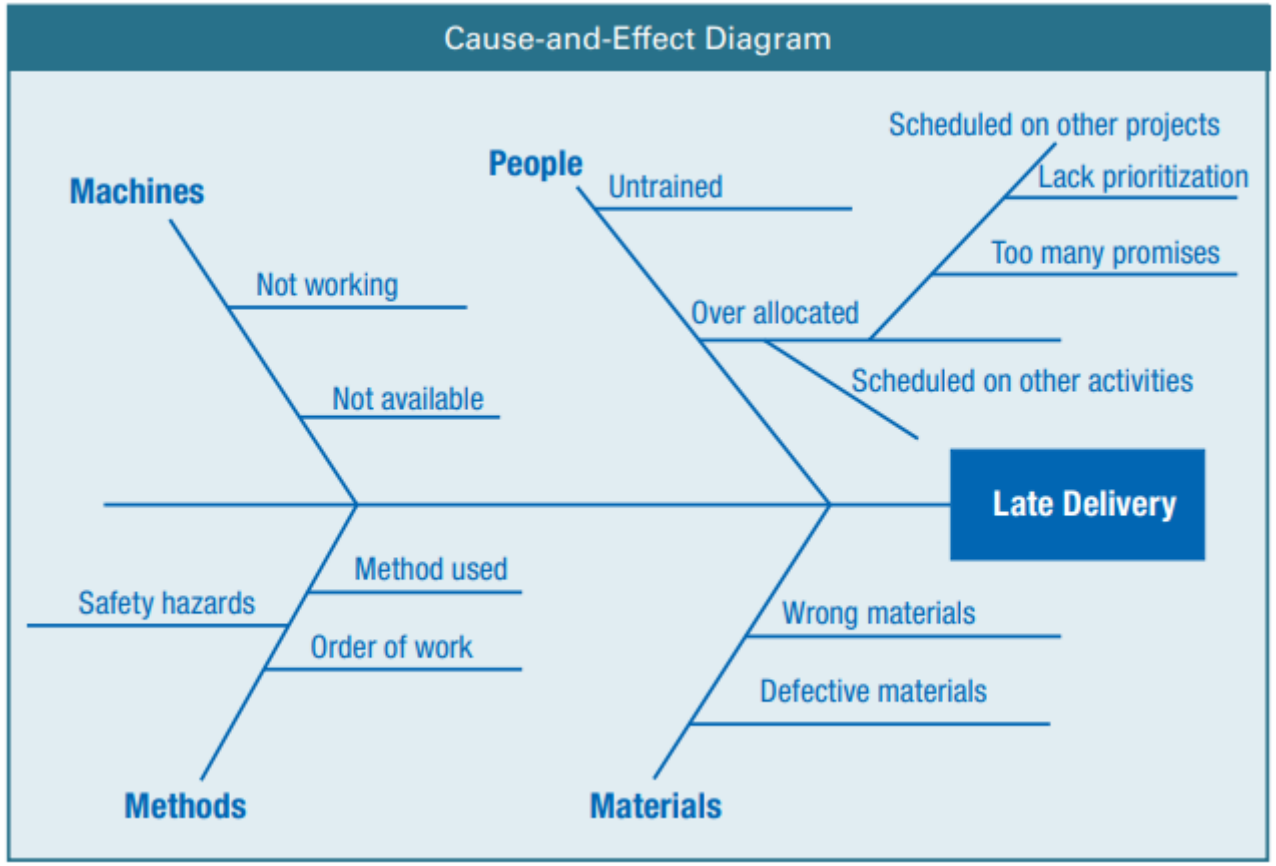
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11.3a Cause-and-Effect Diagram

- List the risk as the effect in a box at the head of the fish.
- Name the big “bones.”
- Complete the smaller “bones.”

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11.3b Perform Quantitative Risk Analysis

Quantitative Risk Analysis is “the process of numerically analyzing the effect of identified risks on overall project objectives.”

- Practice Standard for Project Risk Management (PMI)

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Common Quantitative Risk Analysis Techniques (1 of 2)

Decision Tree Analysis	A graphic tool depicting alternative choices as branches, multiple options for each alternative, and evaluating potential outcomes in terms of uncertainty and monetary value.
Expected Monetary Value (EVA) **Analysis**	A statistical technique to calculate present value of future outcomes to choose the best alternative.
Failure Mode and Effect Analysis (FMEA)	"A step by step approach for identifying all possible failures in a design, a manufacturing or assembly process, or a product or service." - Practice Standard for Project Risk Management (PMI)

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Common Quantitative Risk Analysis Techniques (2 of 2)

Sensitivity Analysis	A quantitative what-if risk analysis technique that presents comparative analyses of various desirable outcomes with respect to a financial measure or uncertainty. It can be used to determine which risks have the most impact on the project outcomes or goals. *(i.e. Tornado diagram)*
Simulation	A technique that mimics real situations using uncertainties and assessing their impact on project objectives. *(i.e. Monte Carlo)*

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Criteria for Selecting a Quantitative Risk Technique Methodology

- Use the explicit knowledge of project team members.
- Allow quick response.
- Help determine project cost and schedule contingency.
- Help foster clear communication.
- Easy to use and understand.

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11.4 Plan Risk Responses (1 of 2)

11.4a Strategies for Responding to Risks
11.4b Risk Register Updates

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11.4 Plan Risk Responses (2 of 2)

Plan risk responses is "the process of developing options and actions to enhance opportunities and reduce threats to project objectives." - Practice Standard for Project Risk Management (PMI)

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11.4a Risk Strategies



- Avoid Risk
- Transfer Risk
- Mitigate Risk
- Accept Risk
- Research Risk
- Exploit Opportunity
- Share Opportunity
- Enhance Opportunity

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Exhibit 11.13

Common Project Risk Strategies

Strategy	Type of Risk	Example
Avoid	Threat	1. Change project plan and/or scope 2. Improve project communications 3. Decide not to perform project
Transfer	Threat	1. Insurance 2. Fixed-price contract 3. Hire expert
Mitigate	Threat	1. Lower probability and/or impact of threat 2. Build in redundancy 3. Use more reliable methods
Accept	Threat and opportunity	1. Deal with it if and when it happens 2. Establish triggers and update frequently 3. Establish time and/or cost contingencies
Research	Threat and opportunity	1. Get more and/or better information 2. Verify assumptions 3. Use prototype
Exploit	Opportunity	1. Assign talented resources to project 2. Give more emphasis to project
Share	Opportunity	1. Allocate partial ownership to third party 2. Form a joint venture
Enhance	Opportunity	1. Increase probability and/or positive impact 2. Identify and maximize key drivers 3. Add more resources

Source: Adapted from Project Management Institute, A *Guide to the Project Management Body of Knowledge* (*PMBOK® Guide*) (Newtown Square, PA: Project Management Institute, 2008): 261–263; Royer, Paul S., *Project Risk Management*: A Proactive *Approach* (Vienna, VA: Management Concepts, Inc., 2002): 35; and Verzuh, Eric, *The Fast Forward MBA in Project Management,* 2nd ed. (Hoboken, NJ: John Wiley & Sons, 2005): 100–103.

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11.4b Risk Register Updates

- Add probability of each risk occurring and its impact to the register.
- Document results of quantitative risk analysis in risk register.

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11.5 *PMBOK Guide 7e*

11.5a Stakeholders
11.5b Team
11.5c Development Approach
11.5d Planning
11.5e Delivery and Measurement
11.5f Uncertainty

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11.5a Stakeholders

- A project team must work collaboratively with stakeholders to elicit their requirements and expectations, then attempt to resolve issues and concerns while developing the Risk Plan.
- Periodic communication with the stakeholders is highly desirable, with a focus on new developments and changes in the Risk Management Plan.

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11.5b Team

- Identifying risks is easier and a more realistic list of risks can be identified when team members possess experience with similar past projects and are knowledgeable about the risks associated with those projects.
- Team members assume a leadership role in monitoring risks during the project execution phase.

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11.5c Development Approach

- When requirements are identified with a greater level of confidence and with less uncertainty, a **predictive development approach** is employed.
- In an **adaptive approach**, uncertainty and ambiguity are often associated with defining requirements.

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11.5d Planning

- In a **predictive planning** approach, develop a detailed WBS and forecast with possible risks associated with each WBS work package.
- In an **adaptive approach**, high-level themes or epics are used to develop features that are further decomposed into stories.

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11.5e Delivery and Measurement

- If uncertainty in defining requirements is minimized, a predictive approach and consequent detailed development of scope will help in identifying, monitoring, and managing risks.
- If uncertainty and volatility are present in requirements, the scope is narrowed down using an adaptive approach so that acceptance criteria, measures, and definition of done can be determined.

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11.5f Uncertainty

- Risks occur when uncertainty exists in defining requirements, and the presence of risk adds to the uncertainty.
- Ambiguity, volatility, and complexity are some of the factors that will increase risks associated with the project.

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11.6 Agile Projects

Plan-Driven versus Agile Comparison on Project Uncertainty Planning
11.6a Agile Terms

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Exhibit 11.15

Comparison of Plan-Driven and Agile Approaches for Project Uncertainty Planning

Project Uncertainty Planning Questions	Plan-Driven	Agile
When are risks identified?	Early if possible, but throughout project	Early if possible, but throughout project
What types of risk analysis are used?	Always qualitative, sometimes quantitative	Always qualitative, sometimes quantitative
What risk response strategies are used?	Avoid, transfer, mitigate, accept, research, exploit, share, enhance	Avoid, transfer, mitigate, accept, research, exploit, share, enhance
What is the project trying to create and how is it delivered?	Outputs by meeting agreements	Outcomes by close coordination with stakeholders
When are risks handled?	Whenever needed	Early by using a risk-based spike and whenever needed
What risks are based upon resources?	Commitments are not honored	There are not enough resources to commit
Where do many lessons learned originate?	Previous projects workers were on	From the long-lived project team

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11.6a Agile Terms

TERM	DEFINITION
Continuously Identify Risks	Specific effort to use feedback and engage team to identify risks.
Premortum	Brainstorming description of what could go wrong.
Assume Variability	Preserve options to provide flexibility because of unknowns.
Handoffs	Whenever work is turned over some tacit knowledge is lost.
Risk-based Spike	Short time-boxed work to address a specific risk.

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Summary

- All projects have some risks.
- Risk planning should use an appropriate level of detail to plan for major risks.
- Risk planning begins with an understanding of project success.
- Risk management plan is part of the overall Project Management Plan.
- Risk identification includes gathering information on potential risks.
- Identified risks are documented in a Risk Register.
- Identified risks are analyzed.
- Risk response planning involves determining response to each **major** risk.
- Risk response strategies include avoid, transfer, mitigate, accept, research, exploit, share, and enhance.

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